Decoding Pragmatic Cues:

Evaluating Audio-Enabled Large Language Models in Detecting Gricean Maxim Flouting

Alexis Liao

Leiden University · e.liao.2@umail.leidenuniv.nl

Audio LLMs · Pragmatics · Gricean maxims · Paralinguistic cues

Background

Paralinguistic features — pitch, stress, timing, filler words, hedges, and voice quality — can alter how an utterance is pragmatically perceived.

These cues may help listeners detect whether a speaker is deliberately flouting a conversational maxim and thereby conveying a conversational implicature. In accordance with Grice, such cues serve as supplementary evidence for pragmatic inference.

Audio-enabled language models can process speech directly, bypassing transcription. However, it remains unclear whether their pragmatic judgments are genuinely sensitive to paralinguistic contrasts or merely reflect model-specific response preferences.

Research Questions

RQ1. Can audio-enabled language models use paralinguistic contrasts to distinguish literal readings from conversational implicatures?

RQ2. Can paired multimodal demonstrations and explicit prosodic guidance improve condition-sensitive, maxim-specific pragmatic reasoning?

We examine Quality, Quantity, and Manner. Relation was excluded because its interpretation depends primarily on discourse context rather than the manipulated paralinguistic contrast.

Methods

Stimuli

- 45 target items: 15 each for Quality, Quantity, and Manner
- Each item included context, a preceding utterance, and matched recordings:
  - Neutral — literal reading expected
  - Marked — implicature and maxim flouting expected
- A transcript-only condition provided a lexical baseline
- Stimuli underwent blind auditing, revision, novelty checking, and maxim relabelling

Models

`Qwen2-Audio-7B-Instruct` · `Kimi-Audio-7B-Instruct` · `Step-Audio-2-mini`

Prompting Regimes

1. Zero-shot
2. Paired 6-shot: neutral and marked recordings from one development item per maxim
3. Prosody-informed 6-shot: the same demonstrations plus explicit, non-deterministic cue–pragmatics guidance

Follow-up demonstrations used three held-out pilot items; all 45 target items remained unseen.

Evaluation

Reading accuracy · Maxim accuracy · Joint accuracy · Correct neutral→marked item pairs · Exact paired tests

Zero-shot Results

Single-condition accuracy was strongly shaped by model-specific response preferences.

- Kimi-Audio predicted a literal reading for 93.3 % of marked trials and selected `none` as the maxim for all 135 trials.
- Qwen2-Audio predicted an implicature for every trial and selected `quality` for almost all audio trials.
- Step-Audio 2 predicted an implicature for every marked trial, but frequently paired `implicature` with `maxim = none`.

Marked-condition reading accuracy was therefore misleadingly high for Qwen2-Audio and Step-Audio 2, since both models also over-predicted implicatures in the neutral condition.

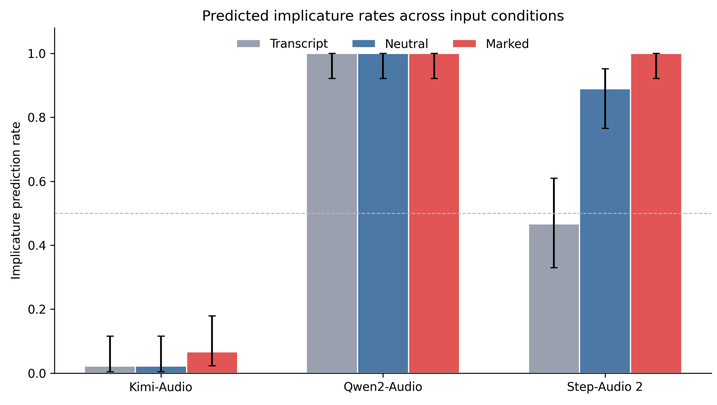
Condition-Sensitive Performance

Model	Neutral acc.	Marked acc.	Correct pairs
Kimi-Audio	97.8 %	6.7 %	2/45 (4.4 %)
Qwen2-Audio	0.0 %	100 %	0/45 (0.0 %)
Step-Audio 2	11.1 %	100 %	5/45 (11.1 %)

No model showed significant zero-shot paired sensitivity:

Kimi: 2/45 correct reading pairs, $p = .50$ · Qwen: 0/45, $p = 1.00$ · Step-Audio: 5/45, $p = .0625$

Implicature Rate by Condition



Zero-shot Failure Profiles

Kimi — **`literal + none`**
Selected `none` for every trial and rarely changed its reading.

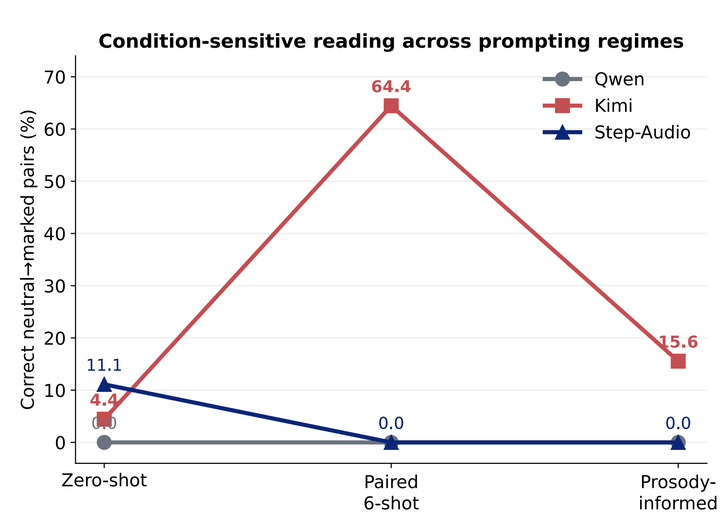
Qwen — **`implicature + quality`**
Applied nearly identical labels across neutral and marked audio.

Step-Audio — **`implicature + none`**
Often predicted implicature without identifying a flouted maxim.

These profiles suggest that implicature detection and maxim identification are separable capabilities.

Prompting Follow-up

We tested whether matched demonstrations and explicit prosodic knowledge could overcome zero-shot response biases.



Kimi-Audio was the only model to benefit from paired demonstrations.

- Correct reading pairs increased from 2/45 to 29/45 ($p < .001$).
- Correct reading + maxim pairs increased from 0/45 to 11/45 ($p < .001$).
- Explicit prosodic guidance reduced reading pairs to 7/45 ($p < .001$ vs. paired 6-shot).

Model-specific Prompt Effects

Qwen remained almost categorically biased toward `implicature + quality` under all three regimes.

Kimi learned the neutral–marked reading contrast from paired demonstrations, but continued to over-select Quality. Explicit guidance shifted it back toward `literal + none`.

Step-Audio copied label alternatives in the paired 6-shot condition, then collapsed to `literal + none` on all 90 prosody-informed trials.

Prompting changed response biases more readily than it improved maxim-specific pragmatic reasoning.

Conclusion

Zero-shot performance did not demonstrate reliable, maxim-specific sensitivity to the neutral–marked contrast. Single-condition accuracy largely reflected model-specific response biases rather than pragmatic competence.

Paired multimodal demonstrations substantially improved Kimi-Audio, increasing correct reading pairs from 2/45 to 29/45. However, maxim identification remained strongly biased toward Quality.

Explicit prosodic guidance provided no additional benefit: Qwen remained unchanged, while Kimi and Step-Audio shifted toward conservative `literal + none` responses.

These findings show that audio access, examples, and explicit cue descriptions do not guarantee robust integration of paralinguistic evidence with pragmatic reasoning.

More explicit prompting changed model biases more readily than it improved pragmatic reasoning.

Limitations: one speaker, no independent human validation, three open-weight models, three maxim categories, and one fixed demonstration set. Future work should add multiple speakers, human judgments, acoustic measurements, and controlled cue manipulations. True fine-tuning is also expected.